

Heavy-tailed distributions

Solutions to the theoretical exercises of Chapter 3, Section 3.8

Notation and standing conventions

Throughout, Z is a non-negative random variable with distribution function F and survival function $\bar{F} = 1 - F$, and we assume $\bar{F}(z) > 0$ for every z (all tail classes below are vacuous or ill-posed for distributions with a finite right endpoint). We write

$$\alpha := 1/\gamma > 0$$

for the tail index whenever this abbreviation shortens formulas, and cst for a finite positive constant whose value may change from line to line and which may depend on fixed parameters (γ , ε , the kernel, ...) but never on the running variables ($z, t, x, y, n, \dots$). Numbering of theorems, conditions and displays refers to the chapter.

The four tail classes of Section 3.1 are, for reference:

$$\begin{aligned} \text{heavy-tailed (HT)} : & \quad \lim_{z \rightarrow \infty} e^{tz} \bar{F}(z) = \infty \quad \text{for all } t > 0, \\ \text{long-tailed (LT)} : & \quad \lim_{z \rightarrow \infty} \bar{F}(z+x)/\bar{F}(z) = 1 \quad \text{for all } x \geq 0, \\ \text{subexponential (SE)} : & \quad \lim_{z \rightarrow \infty} \mathbb{P}(Z_1 + Z_2 > z) / (2\bar{F}(z)) = 1, \\ \text{regularly varying (RV)} : & \quad \bar{F}(z) = z^{-1/\gamma} \ell(z), \quad \ell \text{ slowly varying.} \end{aligned}$$

Exercise 1 $\text{RV} \subset \text{SE} \subset \text{LT} \subset \text{HT}$

Solution. We prove the three inclusions in the order in which they are easiest, starting from the weakest class.

(a) Long-tailed $\Rightarrow$ heavy-tailed. Fix $t > 0$ and set $h(z) := e^{tz} \bar{F}(z) > 0$. Then

$$\frac{h(z+1)}{h(z)} = e^t \frac{\bar{F}(z+1)}{\bar{F}(z)} \xrightarrow{z \rightarrow \infty} e^t > 1,$$

so there is z_0 with $h(z+1) \geq e^{t/2} h(z)$ for all $z \geq z_0$, whence by induction $h(z_0+n) \geq e^{nt/2} h(z_0) \rightarrow \infty$. To upgrade this to a genuine limit, let $z \in [z_0+n, z_0+n+1)$ and use that $\bar{F}$ is non-increasing:

$$h(z) = e^{tz} \bar{F}(z) \geq e^{t(z_0+n)} \bar{F}(z_0+n+1) = e^{-t} h(z_0+n+1) \geq e^{-t} e^{(n+1)t/2} h(z_0).$$

Letting $z \rightarrow \infty$ (hence $n \rightarrow \infty$) gives $h(z) \rightarrow \infty$, which is (3.1). Since $t > 0$ was arbitrary, $\mathbb{E}[e^{tZ}] = \infty$ for every $t > 0$ and the distribution is heavy-tailed.

(b) Subexponential $\Rightarrow$ long-tailed. Conditioning on Z_2 and using $Z_i \geq 0$,

$$\mathbb{P}(Z_1 + Z_2 > z) = \bar{F}(z) + \int_{[0,z]} \bar{F}(z-y) F(dy),$$

so subexponentiality is equivalent to

$$\int_{[0,z]} \frac{\bar{F}(z-y)}{\bar{F}(z)} F(dy) \rightarrow 1, \quad z \rightarrow \infty. \tag{1}$$

Fix $x > 0$ and take $z > x$ large enough that $F(x) < 1$. Splitting the integral and using monotonicity of $\bar{F}$ ($\bar{F}(z-y) \geq \bar{F}(z)$ on $[0, x]$, and $\bar{F}(z-y) \geq \bar{F}(z-x)$ on $(x, z]$),

$$\int_{[0,z]} \frac{\bar{F}(z-y)}{\bar{F}(z)} F(dy) \geq F(x) + \frac{\bar{F}(z-x)}{\bar{F}(z)} (F(z) - F(x)).$$

Letting $z \rightarrow \infty$, the left-hand side tends to 1 by (1) and $F(z) \rightarrow 1$, so

$$\limsup_{z \rightarrow \infty} \frac{\bar{F}(z-x)}{\bar{F}(z)} \leq \frac{1-F(x)}{1-F(x)} = 1.$$

Since $\bar{F}(z-x)/\bar{F}(z) \geq 1$ always, the limit equals 1; replacing z by $z+x$ gives $\bar{F}(z+x)/\bar{F}(z) \rightarrow 1$, i.e. F is long-tailed.

(c) Regularly varying $\Rightarrow$ subexponential. Let $\bar{F}(z) = z^{-\alpha} \ell(z)$ with $\alpha = 1/\gamma > 0$.

Lower bound (valid for any distribution with $\bar{F}(z) \rightarrow 0$). With $M_2 = Z_1 \vee Z_2$,

$$\mathbb{P}(Z_1 + Z_2 > z) \geq \mathbb{P}(M_2 > z) = 2\bar{F}(z) - \bar{F}(z)^2, \quad \text{so} \quad \liminf_{z \rightarrow \infty} \frac{\mathbb{P}(Z_1 + Z_2 > z)}{2\bar{F}(z)} \geq 1.$$

Upper bound. Fix $v \in (0, 1/2)$. If $Z_1 + Z_2 > z$ then either $Z_1 > (1-v)z$, or $Z_2 > (1-v)z$, or both variables lie in $(vz, (1-v)z]$; indeed if both are $\leq (1-v)z$ and their sum exceeds z , then each exceeds $z - (1-v)z = vz$. Hence

$$\mathbb{P}(Z_1 + Z_2 > z) \leq 2\bar{F}((1-v)z) + \bar{F}(vz)^2.$$

Dividing by $\bar{F}(z)$ and using regular variation twice,

$$\frac{2\bar{F}((1-v)z)}{\bar{F}(z)} \rightarrow 2(1-v)^{-\alpha}, \quad \frac{\bar{F}(vz)^2}{\bar{F}(z)} = \frac{\bar{F}(vz)}{\bar{F}(z)} \cdot \bar{F}(vz) \rightarrow v^{-\alpha} \cdot 0 = 0.$$

Therefore $\limsup_z \mathbb{P}(Z_1 + Z_2 > z)/\bar{F}(z) \leq 2(1-v)^{-\alpha}$ for every $v \in (0, 1/2)$, and letting $v \downarrow 0$ gives $\limsup \leq 2$. Combined with the lower bound, (3.2) holds.

Remark 0.1. The upper bound in (c) used only the two facts $\bar{F}(cz)/\bar{F}(z) \rightarrow c^{-\alpha}$ and $\bar{F}(z) \rightarrow 0$; the same “one big jump” splitting is the standard route to subexponentiality for any long-tailed class, and it is what we reuse in Exercise 2(iii).

Exercise 2 Counterexamples to the reversed inclusions

Solution. We exhibit, in turn, a heavy-tailed law that is not long-tailed, a long-tailed law that is not subexponential, and a subexponential law that is not regularly varying.

(i) Heavy-tailed but not long-tailed. Let Z be discrete with

$$\mathbb{P}(Z = 4^n) = 2^{-n}, \quad n \geq 1,$$

so that $\bar{F}(z) = 2^{-n}$ for $z \in [4^n, 4^{n+1})$ and $\bar{F}(z) = 1$ for $z < 4$.

Heavy-tailed: for every $t > 0$, $\mathbb{E}[e^{tZ}] = \sum_{n \geq 1} 2^{-n} e^{t4^n} = \infty$, because $e^{t4^n} 2^{-n} \rightarrow \infty$; the same computation gives $e^{tz} \bar{F}(z) \geq e^{t4^n} 2^{-n} \rightarrow \infty$ for $z \in [4^n, 4^{n+1})$, which is (3.1).

Not long-tailed: take $x = 1$ and $z_n = 4^n - 1$. Then

$$\frac{\bar{F}(z_n + 1)}{\bar{F}(z_n)} = \frac{\bar{F}(4^n)}{\bar{F}(4^n - 1)} = \frac{2^{-n}}{2^{-(n-1)}} = \frac{1}{2} \not\rightarrow 1.$$

The tail is heavy, but it is not asymptotically translation invariant: the mass sits on a sparse lattice, and a shift of size 1 can still “see” an atom.

(ii) Long-tailed but not subexponential. The idea is to build a tail $\bar{F} = e^{-H}$ whose hazard H has vanishing slope (which forces long tails) but alternates between long flat stretches and steep stretches, so that the convolution tail at $2c$ is dominated by the event that *both* summands are of size $\approx c$.

Choose $c_1 \geq 4$ and $c_{n+1} \geq c_n^2$, and define $H : [0, \infty) \rightarrow [0, \infty)$ by $H \equiv 0$ on $[0, c_1]$ and, recursively for $n \geq 1$,

$$H(z) = H(c_n) + \frac{z - c_n}{\sqrt{c_n}} \quad \text{for } z \in [c_n, 2c_n], \quad H(z) = H(2c_n) \quad \text{for } z \in [2c_n, c_{n+1}].$$

Thus H is continuous, non-decreasing, $H(c_n) = S_n := \sum_{j < n} \sqrt{c_j}$ and $H(2c_n) = S_n + \sqrt{c_n}$. Since $c_{j+1} \geq c_j^2 \geq 4c_j$, we have $S_n \leq 2\sqrt{c_{n-1}}$. Set $\bar{F} := e^{-H}$, a genuine survival function ($H(0) = 0$, $H \uparrow \infty$).

Long-tailed: the slope of H on $[c_n, \infty)$ is at most $1/\sqrt{c_n}$, so for fixed $x \geq 0$ and $z \geq c_n$,

$$1 \geq \frac{\bar{F}(z+x)}{\bar{F}(z)} = e^{-(H(z+x)-H(z))} \geq e^{-x/\sqrt{c_n}} \xrightarrow{n \rightarrow \infty} 1.$$

(In particular, by Exercise 1(a), F is heavy-tailed.)

Not subexponential: evaluate at $z_n = 2c_n$. On the one hand $\bar{F}(2c_n) = e^{-S_n - \sqrt{c_n}}$; on the other hand, by independence,

$$\mathbb{P}(Z_1 + Z_2 > 2c_n) \geq \mathbb{P}(Z_1 > c_n)\mathbb{P}(Z_2 > c_n) = e^{-2S_n}.$$

Hence

$$\frac{\mathbb{P}(Z_1 + Z_2 > 2c_n)}{2\bar{F}(2c_n)} \geq \frac{1}{2} e^{\sqrt{c_n} - S_n} \geq \frac{1}{2} e^{\sqrt{c_n} - 2\sqrt{c_{n-1}}} \rightarrow \infty,$$

because $\sqrt{c_n} \geq c_{n-1} \gg 2\sqrt{c_{n-1}}$. So (3.2) fails along (z_n) .

(iii) Subexponential but not regularly varying. Take the Weibull tail of Exercise 4 with shape $a \in (0, 1)$, i.e. $\bar{F}(z) = e^{-z^a}$, $z \geq 0$. Exercise 4(b) shows it is not regularly varying (for any $a > 0$). We check subexponentiality. Write $f(y) = ay^{a-1}e^{-y^a}$ and split the event $\{Z_1 + Z_2 > z\}$ according to which summand is at most $z/2$:

$$\mathbb{P}(Z_1 + Z_2 > z) = 2 \int_{[0, z/2]} \bar{F}(z-y) f(y) dy + \mathbb{P}(Z_1 > z/2, Z_2 > z/2, Z_1 + Z_2 > z).$$

The last term is at most $\bar{F}(z/2)^2 = e^{-2^{1-a}z^a}$, and $\bar{F}(z/2)^2/\bar{F}(z) = e^{-(2^{1-a}-1)z^a} \rightarrow 0$ since $a < 1$. It therefore suffices to prove

$$I_z := \int_0^{z/2} \frac{\bar{F}(z-y)}{\bar{F}(z)} f(y) dy \rightarrow 1.$$

Because $y \mapsto y^a$ is concave with value 0 at the origin, it is subadditive, so $z^a - (z-y)^a - y^a \leq 0$ and the integrand is bounded by $f(y)e^{y^a} \cdot e^0$; more usefully, for $y \leq z/2$ the mean value theorem gives $z^a - (z-y)^a \leq ay(z/2)^{a-1}$.

- On $[0, A]$ with A fixed: the exponent is at most $aA(z/2)^{a-1} \rightarrow 0$, so this part converges to $F(A)$.
- On $[A, u_0z]$: writing $y = uz$, the exponent equals $z^a\varphi(u)$ with $\varphi(u) = 1 - (1-u)^a - u^a$, and since $1 - (1-u)^a \leq 2^{1-a}au$ for $u \leq 1/2$, we have $\varphi(u) \leq -u^a(1 - 2^{1-a}au^{1-a}) \leq -u^a/2$ for $u \leq u_0$ small enough. Hence the integrand is bounded by $ay^{a-1}e^{-y^a/2}$, whose integral over $[A, \infty)$ is $2e^{-A^a/2}$, small uniformly in z for large A .

- On $[u_0 z, z/2]$: φ is continuous and strictly negative on $[u_0, 1/2]$, so $\varphi \leq -\delta < 0$ there and the contribution is at most $e^{-\delta z^a} \rightarrow 0$.

Letting $z \rightarrow \infty$ and then $A \rightarrow \infty$ yields $I_z \rightarrow 1$, and therefore $\mathbb{P}(Z_1 + Z_2 > z)/(2\bar{F}(z)) \rightarrow 1$.

Remark 0.2. For (iii) one may equally take the lognormal law of Example 3.2, which is heavy-tailed and not regularly varying; the general principle behind both examples is the classical criterion: if $\bar{F} = e^{-H}$ with H eventually concave and $H(z)/z \rightarrow 0$, then F is subexponential.

Exercise 3 $\gamma = \int_1^\infty T(s) s^{-1} ds$; sum versus integral Hill

Solution. The identity. With $T(s) = s^{-1/\gamma}$ and $\gamma > 0$,

$$\int_1^\infty \frac{T(s)}{s} ds = \int_1^\infty s^{-1/\gamma-1} ds = \left[-\gamma s^{-1/\gamma} \right]_1^\infty = \gamma.$$

Probabilistically: if Z is exactly Pareto above the threshold u , then $Z/u \mid Z > u$ has survival function T , so $\int_1^\infty T(s) s^{-1} ds = \mathbb{E}[\log(Z/u) \mid Z > u] = \gamma$, which is precisely the “mean of a log-Pareto” motivation for the Hill estimator.

Sum version = integral version. Let $\hat{T}_n(s) = k^{-1} \sum_{i=1}^n \mathbf{1}(Z_i > Z_{n,n-k}s)$ be the tail empirical function with random levels (Definition 3.6), and assume the Z_i are distinct (a.s. true for continuous F). By Tonelli,

$$\int_1^\infty \frac{\hat{T}_n(s)}{s} ds = \frac{1}{k} \sum_{i=1}^n \int_1^\infty \mathbf{1}\left(s < \frac{Z_i}{Z_{n,n-k}}\right) \frac{ds}{s} = \frac{1}{k} \sum_{i=1}^n \log^+\left(\frac{Z_i}{Z_{n,n-k}}\right).$$

Exactly the k largest observations exceed $Z_{n,n-k}$, i.e. the summand is non-zero precisely for $Z_i \in \{Z_{n,n}, \dots, Z_{n,n-k+1}\}$, so

$$\int_1^\infty \frac{\hat{T}_n(s)}{s} ds = \frac{1}{k} \sum_{i=1}^k \log \frac{Z_{n,n-i+1}}{Z_{n,n-k}} = \gamma_{n,k},$$

which is Definition 3.2. Thus the Hill estimator is the empirical version of $\gamma = \int_1^\infty T(s) s^{-1} ds$ obtained by substituting $\hat{T}_n$ for T : the sum form is a plug-in of order statistics, the integral form is a linear functional of the tail empirical process. The integral form is the useful one for the asymptotics, because weak convergence of $\hat{T}_n$ (Theorems 3.10–3.12) transfers to $\gamma_{n,k}$ by continuity of $g \mapsto \int_1^\infty g(s) s^{-1} ds$, once the tail $\int_m^\infty$ has been shown negligible.

Exercise 4 Weibull tails

Let $\bar{F}(x) = \exp(-x^a)$, $x \geq 0$, $a > 0$, with density $f(x) = a x^{a-1} e^{-x^a}$.

(a) **Finiteness of the m.g.f., and heavy-tailedness iff $a < 1$**

Solution. For $t \leq 0$ we have $M(t) = \mathbb{E}[e^{tZ}] \leq 1 < \infty$ since $Z \geq 0$, so only $t > 0$ matters. Using $e^{tZ} = 1 + \int_0^Z t e^{tx} dx$ and Tonelli,

$$M(t) = \mathbb{E}[e^{tZ}] = 1 + t \int_0^\infty e^{tx} \bar{F}(x) dx = 1 + t \int_0^\infty e^{tx-x^a} dx,$$

so for $t > 0$, $M(t) < \infty$ if and only if $\int_0^\infty e^{tx-x^a} dx < \infty$; everything depends on the sign of $tx - x^a$ for large x .

Case $a > 1$. Fix $t > 0$. Since $x^a/x = x^{a-1} \rightarrow \infty$, for $x \geq x_t := (2t)^{1/(a-1)}$ we get $x^a \geq 2tx$ and hence $tx - x^a \leq -tx$. Thus $\int_{x_t}^{\infty} e^{tx-x^a} dx \leq \int_{x_t}^{\infty} e^{-tx} dx < \infty$, while the integral over the compact $[0, x_t]$ is finite. Hence $M(t) < \infty$ for every $t > 0$.

Case $a = 1$ (exponential). $\int_0^{\infty} e^{(t-1)x} dx < \infty$ iff $t < 1$, so $M(t) = (1-t)^{-1} < \infty$ for $t < 1$ and $M(t) = \infty$ for $t \geq 1$. In particular M is finite on a neighbourhood of the origin.

Case $a < 1$. Fix $t > 0$. Now $x^a = o(x)$, so with $x_t := (2/t)^{1/(1-a)}$ we have $x^a \leq \frac{t}{2}x$, hence $tx - x^a \geq \frac{t}{2}x$, for all $x \geq x_t$. Therefore $\int_0^{\infty} e^{tx-x^a} dx \geq \int_{x_t}^{\infty} e^{tx/2} dx = \infty$ and $M(t) = \infty$ for every $t > 0$.

Collecting the three cases: $M(t) < \infty$ for some $t > 0 \iff a \geq 1$. Consequently M is finite in a neighbourhood $(-\varepsilon, \varepsilon)$ of the origin iff $a \geq 1$, i.e. by the definition in Section 3.1 the Weibull distribution is heavy-tailed if and only if $a < 1$. Equivalently, in the form (3.1): $e^{tx}\bar{F}(x) = e^{tx-x^a} \rightarrow \infty$ for every $t > 0$ iff $a < 1$.

Remark 0.3. For every $a > 0$ all moments are finite: substituting $v = x^a$,

$$\mathbb{E}[Z^k] = \int_0^{\infty} kx^{k-1}\bar{F}(x) dx = \Gamma(1+k/a) < \infty, \quad k > 0.$$

So for $a < 1$ the Weibull law has all moments finite yet no moment generating function on any interval $(0, \varepsilon)$: no Chernoff bound $\mathbb{P}(Z > x) \leq e^{-tx}M(t)$ is available, and sums obey the single-big-jump principle rather than exponential concentration.

(b) $\bar{F}(tx)/\bar{F}(x) \rightarrow 0$ and failure of regular variation

Solution. For $t > 1$ and $a > 0$,

$$\frac{\bar{F}(tx)}{\bar{F}(x)} = \exp(x^a - (tx)^a) = \exp(-x^a(t^a - 1)) \xrightarrow{x \rightarrow \infty} 0,$$

since $t^a - 1 > 0$. If $\bar{F}$ were regularly varying with index $1/\gamma \geq 0$, i.e. $\bar{F}(x) = x^{-1/\gamma}\ell(x)$ with ℓ slowly varying, we would have

$$\frac{\bar{F}(tx)}{\bar{F}(x)} = t^{-1/\gamma} \frac{\ell(tx)}{\ell(x)} \rightarrow t^{-1/\gamma} \in (0, 1],$$

a strictly positive limit, contradicting the limit 0 just obtained. Hence the Weibull law is not regularly varying for *any* $a > 0$.

Together with (a) and Exercise 2(iii): for $a \in (0, 1)$ the Weibull distribution is heavy-tailed and even subexponential, yet lies strictly outside the regularly varying class; its tail decays faster than any power but slower than any exponential. It belongs to the Gumbel max-domain of attraction, consistent with item 2 of the list in Section 3.2.1.

Exercise 5 The Burr tail is regularly varying

Solution. Let $F(x) = 1 - (1 + (x/\theta)^\eta)^{-\beta}$ for $x \geq 0$, with $\theta, \eta, \beta > 0$, so that $\bar{F}(x) = (1 + (x/\theta)^\eta)^{-\beta}$. Factor out the leading power:

$$\bar{F}(x) = \left(\frac{x}{\theta}\right)^{-\eta\beta} \left(1 + \left(\frac{\theta}{x}\right)^\eta\right)^{-\beta} = x^{-\eta\beta} \ell(x), \quad \ell(x) := \theta^{\eta\beta} \left(1 + \left(\frac{\theta}{x}\right)^\eta\right)^{-\beta}.$$

Since $\ell(x) \rightarrow \theta^{\eta\beta} \in (0, \infty)$ as $x \rightarrow \infty$, for every $t > 0$ we get $\ell(tx)/\ell(x) \rightarrow 1$, i.e. ℓ is slowly varying. Hence $\bar{F}$ is regularly varying with index

$$\frac{1}{\gamma} = \eta\beta, \quad \text{equivalently} \quad \boxed{\gamma = \frac{1}{\eta\beta}}.$$

Directly, one may also verify $\overline{F}(tx)/\overline{F}(x) = [(1 + (tx/\theta)^\eta)/(1 + (x/\theta)^\eta)]^{-\beta} \rightarrow t^{-\eta\beta}$.

By Theorem 3.2 the Burr distribution lies in the Fréchet max-domain of attraction with extreme-value index $\gamma = (\eta\beta)^{-1}$; this is the value $\gamma = 1/2$ used with $\eta = \beta = \sqrt{2}$ in Numeric examples 3.1–3.2 and in the practical exercises.

Exercise 6 Potter's bounds

(a) Discussion of the proof of Theorem 3.5

Solution. Write $\ell(z) = z^{1/\gamma}\overline{F}(z)$, which is slowly varying by assumption. The proof has four ingredients; we spell them out and point at the places where care is needed.

(1) *The key analytic fact: a slowly varying function is squeezed between any two powers.* The proof asserts $z^{-\varepsilon}\ell(z) \rightarrow 0$ and $z^\varepsilon\ell(z) \rightarrow \infty$ because these are regularly varying with indices $-\varepsilon$ and $+\varepsilon$. This deserves a proof, and it follows from the representation Theorem 3.7 (applied to ℓ): there are z_0 and functions $c(\cdot) \rightarrow c_0 \in (0, \infty)$, $\delta(\cdot) \rightarrow 0$ with

$$\ell(z) = c(z) \exp\left(\int_{z_0}^z \frac{\delta(u)}{u} du\right), \quad z \geq z_0.$$

Given $\varepsilon > 0$, pick $z_1 \geq z_0$ with $|\delta(u)| \leq \varepsilon/2$ for $u \geq z_1$; then for $z \geq z_1$,

$$\text{cst } z^{-\varepsilon/2} \leq \ell(z) \leq \text{cst } z^{\varepsilon/2}, \quad (2)$$

whence $z^{-\varepsilon}\ell(z) \leq \text{cst } z^{-\varepsilon/2} \rightarrow 0$ and $z^\varepsilon\ell(z) \geq \text{cst } z^{\varepsilon/2} \rightarrow \infty$. This is the only analytic input; everything else is bookkeeping.

(2) *From a limit to a uniform bound.* $z^{-\varepsilon}\ell(z) \rightarrow 0$ gives $\sup_{z \geq b} z^{-\varepsilon}\ell(z) < \infty$ for some b , but the claim of Theorem 3.5 is for *all* $z > 0$, so the supremum must be controlled on $(0, b)$ as well. The proof does this in two pieces:

- on $[1, b]$: $z^{-\varepsilon}\ell(z) = z^{1/\gamma-\varepsilon}\overline{F}(z)$ is bounded because $\overline{F} \leq 1$ and the power is bounded on a compact;
- on $(0, 1)$: $z^{-\varepsilon}\ell(z) = z^{1/\gamma-\varepsilon}\overline{F}(z) \leq 1$, using $\overline{F} \leq 1$ and $\varepsilon < 1/\gamma$ so that the exponent $1/\gamma - \varepsilon$ is positive.

This is exactly where the restriction $\varepsilon \in (0, 1/\gamma)$ is used; without it the first bound would fail near the origin. Combining, $C_\varepsilon := \sup_{z > 0} z^{-\varepsilon}\ell(z) < \infty$ and

$$\overline{F}(z) = z^{-1/\gamma+\varepsilon} \cdot z^{-\varepsilon}\ell(z) \leq C_\varepsilon z^{-1/\gamma+\varepsilon}, \quad z > 0.$$

(3) *The lower bound and why it cannot hold near 0.* Here one needs $\inf_{z \geq \varepsilon} z^\varepsilon\ell(z) > 0$. The text's phrase “strictly positive and diverging to infinity” is slightly loose: divergence gives positivity of the infimum only outside a compact, and on a compact $[\varepsilon, b]$ one should invoke monotonicity of $\overline{F}$,

$$z^\varepsilon\ell(z) = z^{\varepsilon+1/\gamma}\overline{F}(z) \geq \varepsilon^{\varepsilon+1/\gamma}\overline{F}(b) > 0, \quad z \in [\varepsilon, b],$$

which is where $\overline{F} > 0$ everywhere is used. Note also that the lower bound *must* exclude a neighbourhood of the origin: $\overline{F}(z) \leq 1$ while $z^{-1/\gamma-\varepsilon} \rightarrow \infty$ as $z \downarrow 0$, so no inequality $\overline{F}(z) \geq \text{cst } z^{-1/\gamma-\varepsilon}$ can hold on all of $(0, \infty)$ with a fixed constant. The threshold “ $z > \varepsilon$ ” in the statement is an arbitrary but convenient choice; any fixed positive threshold works, with a constant depending on it.

(4) *What the bounds do and do not say.* The constants blow up as $\varepsilon \downarrow 0$ (in (2) the point z_1 moves out to infinity), so Potter's bounds are not a substitute for regular variation itself; their role is to provide integrable majorants that are uniform in n , which is exactly how they are used in the proofs of Theorems 3.13, 3.14 and 3.21, and in Exercise 7 below. Finally, the first bound shows directly that regular variation implies heavy tails: $e^{tz}\overline{F}(z) \geq \text{cst } e^{tz} z^{-1/\gamma-\varepsilon} \rightarrow \infty$ for z large, which is the “general case” alluded to in Section 3.1.

(b) Proof of Theorem 3.6 (Potter's bounds, second version)

Solution. Claim. For every $\varepsilon \in (0, 1/\gamma)$ there is a finite constant C such that

$$\frac{\overline{F}(x/y)}{\overline{F}(x)} \leq C \left(y^{1/\gamma-\varepsilon} \mathbf{1}_{\{y < 1\}} + y^{1/\gamma+\varepsilon} \mathbf{1}_{\{y \geq 1\}} \right) = C(y^{1/\gamma-\varepsilon} \vee y^{1/\gamma+\varepsilon}), \quad x \geq 1, y > 0. \quad (3)$$

(The two expressions agree because $y^{1/\gamma-\varepsilon} \geq y^{1/\gamma+\varepsilon}$ exactly when $y \leq 1$.)

Step 1: a two-sided ratio bound far out. By Theorem 3.7, $\overline{F}(z) = c(z) \exp(\int_{z_0}^z \delta(u) u^{-1} du)$ with $c(z) \rightarrow c_0 \in (0, \infty)$ and $\delta(z) \rightarrow -1/\gamma$. Fix $\varepsilon > 0$ and choose $z_1 \geq z_0$ such that $|\delta(u) + 1/\gamma| \leq \varepsilon$ for $u \geq z_1$ and such that $A := \sup_{v, w \geq z_1} c(w)/c(v) < \infty$ (possible since $c \rightarrow c_0 \in (0, \infty)$). For $w, z \geq z_1$,

$$\frac{\overline{F}(w)}{\overline{F}(z)} = \frac{c(w)}{c(z)} \exp\left(-\int_w^z \frac{\delta(u)}{u} du\right).$$

If $w \leq z$ the exponent is at most $(1/\gamma + \varepsilon) \log(z/w)$; if $w \geq z$ it is at most $-(1/\gamma - \varepsilon) \log(w/z)$. In both cases

$$\frac{\overline{F}(w)}{\overline{F}(z)} \leq A \left(\left(\frac{z}{w}\right)^{1/\gamma+\varepsilon} \vee \left(\frac{z}{w}\right)^{1/\gamma-\varepsilon} \right), \quad w, z \geq z_1. \quad (4)$$

Step 2: case analysis. Put $w = x/y$, $z = x$, so $z/w = y$, and recall from Theorem 3.5 that $\overline{F}(v) \leq C_1 v^{-1/\gamma+\varepsilon}$ for all $v > 0$ and $\overline{F}(v) \geq C_2 v^{-1/\gamma-\varepsilon}$ for $v \geq 1$.

- $x \geq z_1$ and $x/y \geq z_1$. Then (4) is exactly (3) with $C = A$.
- $x/y < z_1$ and $y \geq 1$. Then $\overline{F}(x/y) \leq 1$ and $x < z_1 y$, so

$$\frac{\overline{F}(x/y)}{\overline{F}(x)} \leq \frac{1}{C_2} x^{1/\gamma+\varepsilon} \leq \frac{z_1^{1/\gamma+\varepsilon}}{C_2} y^{1/\gamma+\varepsilon}.$$

- $1 \leq x < z_1$. Then $\overline{F}(x) \geq \overline{F}(z_1) > 0$. If $y < 1$, using $x \geq 1$ and $-1/\gamma + \varepsilon < 0$,

$$\frac{\overline{F}(x/y)}{\overline{F}(x)} \leq \frac{C_1}{\overline{F}(z_1)} \left(\frac{x}{y}\right)^{-1/\gamma+\varepsilon} = \frac{C_1}{\overline{F}(z_1)} x^{-1/\gamma+\varepsilon} y^{1/\gamma-\varepsilon} \leq \frac{C_1}{\overline{F}(z_1)} y^{1/\gamma-\varepsilon};$$

if $y \geq 1$ then simply $\overline{F}(x/y)/\overline{F}(x) \leq 1/\overline{F}(z_1) \leq \overline{F}(z_1)^{-1} y^{1/\gamma+\varepsilon}$.

These three cases are exhaustive: the only combination not listed is $x \geq z_1$, $x/y < z_1$, $y < 1$, which is impossible because $y < 1$ forces $x/y > x \geq z_1$. Taking C to be the largest of the constants produced gives (3). $\square$

How it is used. With $x = u_n$ and $y = 1/t \leq 1$ one gets the bound used repeatedly in the chapter,

$$T_n(t) = \frac{\overline{F}(u_n t)}{\overline{F}(u_n)} \leq \text{cst } t^{-1/\gamma+\varepsilon}, \quad t \geq 1, u_n \geq 1,$$

uniformly in n : this is the integrable majorant that makes the tail $\int_m^\infty$ negligible in Theorems 3.13 and 3.14.

Exercise 7 Breiman's lemma

Solution. Let Z be regularly varying with index $\alpha = 1/\gamma > 0$, let $W \geq 0$ be independent of Z with $\mathbb{E}[W^{\alpha+\varepsilon}] < \infty$ for some $\varepsilon > 0$, and assume $\mathbb{P}(W > 0) > 0$ (otherwise $ZW \equiv 0$ and the statement is empty). Note first that $\mathbb{E}[W^\alpha] \leq 1 + \mathbb{E}[W^{\alpha+\varepsilon}] < \infty$, and $\mathbb{E}[W^\alpha] > 0$.

Step 1: conditioning. By independence, for $x > 0$,

$$\frac{\mathbb{P}(ZW > x)}{\mathbb{P}(Z > x)} = \mathbb{E} \left[\frac{\bar{F}_Z(x/W)}{\bar{F}_Z(x)} \right],$$

with the convention that the integrand is 0 on $\{W = 0\}$ (there $ZW = 0 < x$), consistent with $w^\alpha|_{w=0} = 0$ since $\alpha > 0$.

Step 2: pointwise convergence. For each fixed $w > 0$, regular variation applied with $t = 1/w$ gives

$$\frac{\bar{F}_Z(x/w)}{\bar{F}_Z(x)} \xrightarrow{x \rightarrow \infty} w^\alpha.$$

Step 3: domination. Choose $\varepsilon' \in (0, \varepsilon)$ with also $\varepsilon' < \alpha$. By Theorem 3.6 (proved in Exercise 6) there is a constant C with, for all $x \geq 1$ and all $w > 0$,

$$\frac{\bar{F}_Z(x/w)}{\bar{F}_Z(x)} \leq C \left(w^{\alpha-\varepsilon'} \mathbf{1}_{\{w < 1\}} + w^{\alpha+\varepsilon'} \mathbf{1}_{\{w \geq 1\}} \right) \leq C(1 + W^{\alpha+\varepsilon'}) \Big|_{W=w},$$

because $w^{\alpha-\varepsilon'} \leq 1$ for $w < 1$. The dominating variable is integrable: $\mathbb{E}[W^{\alpha+\varepsilon'}] \leq 1 + \mathbb{E}[W^{\alpha+\varepsilon}] < \infty$. This is precisely where the *extra* moment $\varepsilon > 0$ is needed; $\mathbb{E}[W^\alpha] < \infty$ alone would not supply an integrable majorant.

Step 4: dominated convergence. Steps 2–3 give

$$\lim_{x \rightarrow \infty} \frac{\mathbb{P}(ZW > x)}{\mathbb{P}(Z > x)} = \mathbb{E}[W^\alpha] = \mathbb{E}[W^{1/\gamma}] \in (0, \infty),$$

which is the stated limit.

Step 5: ZW is regularly varying with the same index. For $t > 0$, write

$$\frac{\mathbb{P}(ZW > tx)}{\mathbb{P}(ZW > x)} = \underbrace{\frac{\mathbb{P}(ZW > tx)}{\mathbb{P}(Z > tx)}}_{\rightarrow \mathbb{E}[W^\alpha]} \cdot \underbrace{\frac{\mathbb{P}(Z > tx)}{\mathbb{P}(Z > x)}}_{\rightarrow t^{-\alpha}} \cdot \underbrace{\frac{\mathbb{P}(Z > x)}{\mathbb{P}(ZW > x)}}_{\rightarrow 1/\mathbb{E}[W^\alpha]} \rightarrow t^{-\alpha},$$

the cancellation being legitimate because $\mathbb{E}[W^\alpha] \in (0, \infty)$. Hence ZW is regularly varying with index $\alpha = 1/\gamma$; explicitly, if $\bar{F}_Z(x) = x^{-\alpha} \ell(x)$ then $\mathbb{P}(ZW > x) = x^{-\alpha} \ell(x) \mathbb{E}[W^\alpha] (1 + o(1))$, so the random scaling changes only the constant in the slowly varying part, not the index.

Remark 0.4. The result explains why regular variation is so robust in insurance modelling: multiplying a heavy-tailed claim size by an independent, lighter inflation/exposure factor W does not change the tail index, only the scale. The moment condition cannot be dropped: if $\mathbb{E}[W^\alpha] = \infty$ the product may have a strictly heavier tail.

Exercise 8 Variance of the weak limit of the Hill estimator

Solution. Step 1: the three representations agree. Let W be standard Brownian motion, $T(t) = t^{-1/\gamma}$, and $B(u) := W(u) - uW(1)$, $u \in [0, 1]$, a standard Brownian bridge. Since $\int_1^\infty T(t)t^{-1}dt = \gamma$ (Exercise 3),

$$\int_1^\infty \frac{B(T(t))}{t} dt = \int_1^\infty \frac{W(T(t))}{t} dt - W(1) \int_1^\infty \frac{T(t)}{t} dt = \int_1^\infty \frac{W(T(t))}{t} dt - \gamma W(1).$$

Substituting $u = T(t) = t^{-1/\gamma}$, i.e. $t = u^{-\gamma}$ and $dt/t = -\gamma du/u$, with $t : 1 \rightarrow \infty$ corresponding to $u : 1 \rightarrow 0$,

$$\int_1^\infty \frac{B(T(t))}{t} dt = \gamma \int_0^1 \frac{B(u)}{u} du.$$

So all three expressions in the statement coincide; call the common value ξ .

Step 2: ξ is well defined in L^2 . Since $\mathbb{E}|B(u)| = \sqrt{2u(1-u)/\pi}$,

$$\mathbb{E} \int_0^1 \frac{|B(u)|}{u} du = \sqrt{2/\pi} \int_0^1 u^{-1/2} \sqrt{1-u} du < \infty,$$

so the integral converges absolutely a.s. and Fubini may be applied below.

Step 3: the variance. Using $\text{Cov}(B(s), B(t)) = s \wedge t - st$ and $\frac{s \wedge t}{st} = \frac{1}{s \vee t} = s^{-1} \wedge t^{-1}$,

$$\text{Var}(\xi) = \gamma^2 \int_0^1 \int_0^1 \frac{\text{Cov}(B(s), B(t))}{st} ds dt = \gamma^2 \int_0^1 \int_0^1 (s^{-1} \wedge t^{-1} - 1) ds dt,$$

which is the expression in the statement. Evaluating it, by symmetry,

$$\int_0^1 \int_0^1 \frac{ds dt}{s \vee t} = 2 \int_0^1 \int_0^t \frac{ds dt}{t} = 2 \int_0^1 1 dt = 2, \quad \int_0^1 \int_0^1 1 ds dt = 1,$$

so $\text{Var}(\xi) = \gamma^2(2 - 1) = \gamma^2$. Since ξ is a centred Gaussian variable (an L^2 -limit of linear functionals of a Gaussian process), $\xi \sim N(0, \gamma^2)$, which is the limit law in Theorem 3.14.

Remark 0.5. The computation makes the bias-variance structure transparent: the term $\int_1^\infty W(T(t))t^{-1}dt$ alone has variance $2\gamma^2$ (first display above), and the correction $-\gamma W(1)$, which comes from estimating the deterministic threshold u_n by the random $Z_{n,n-k}$, removes exactly half of it. Estimating the threshold *reduces* the asymptotic variance.

Exercise 9 Consistency of the moment estimator, $\gamma > 0$

Solution. Write $q := Z_{n,n-k}$, $u := u_n$, $r := q/u$, and recall $M_n^{(r)} = k^{-1} \sum_{i=1}^k \log^r(Z_{n,n-i+1}/Z_{n,n-k})$. By Exercise 3, $M_n^{(1)} = \gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma$ (Theorem 3.13), so only $M_n^{(2)}$ requires work; we show $M_n^{(2)} \xrightarrow{\mathbb{P}} 2\gamma^2$ and then conclude by the continuous mapping theorem.

Step 1: integral representation. As in Exercise 3, $\hat{T}_n(t) = k^{-1} \sum_i \mathbf{1}(Z_i > qt)$ is a non-increasing step function with $\hat{T}_n(t) = 0$ for $t \geq Z_{n,n}/q$. Integrating by parts (the boundary terms vanish because $\log^2 1 = 0$ and $\hat{T}_n$ has bounded support),

$$M_n^{(2)} = - \int_1^\infty \log^2(t) \hat{T}_n(dt) = 2 \int_1^\infty \frac{\log t}{t} \hat{T}_n(t) dt.$$

Step 2: passing to deterministic levels. Since $\hat{T}_n(t) = \tilde{T}_n(tr)$, the substitution $v = tr$ (so $dt/t = dv/v$, $\log t = \log v - \log r$) gives

$$M_n^{(2)} = 2 \int_r^1 \frac{\log v}{v} \tilde{T}_n(v) dv + 2 \int_1^\infty \frac{\log v}{v} \tilde{T}_n(v) dv - 2 \log(r) \int_r^\infty \frac{\tilde{T}_n(v)}{v} dv. \quad (5)$$

(This is the decomposition displayed in the proof of Proposition 3.2; note that the sign of the third term there should be a minus, which is immaterial since the term is $o_{\mathbb{P}}(1)$.)

Step 3: the first and third terms are $o_{\mathbb{P}}(1)$. By Theorem 3.9, $r = q/u \xrightarrow{\mathbb{P}} 1$, so we may assume $r \in (1 - \varepsilon, 1 + \varepsilon)$. For the third term, $\int_r^\infty \tilde{T}_n(v)v^{-1}dv = \gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma = O_{\mathbb{P}}(1)$ while $\log r \xrightarrow{\mathbb{P}} 0$, so the product vanishes in probability. For the first term,

$$\left| \int_r^1 \frac{\log v}{v} \tilde{T}_n(v) dv \right| \leq \frac{|\log r| |1 - r|}{1 - \varepsilon} \sup_{v \in (1 - \varepsilon, 1 + \varepsilon)} \tilde{T}_n(v) = o_{\mathbb{P}}(1),$$

since $\tilde{T}_n(v) \leq \tilde{T}_n(1 - \varepsilon) = O_{\mathbb{P}}(1)$ by Theorem 3.9.

Step 4: the main term on compacts. By uniform consistency of the tail empirical function on compacts bounded away from 0 (Theorem 3.9), for every $m > 1$

$$\gamma_n^m := 2 \int_1^m \frac{\log v}{v} \tilde{T}_n(v) dv \xrightarrow{\mathbb{P}} 2 \int_1^m \frac{\log v}{v} v^{-1/\gamma} dv =: \gamma^m,$$

and, computing the full integral with $\alpha = 1/\gamma$,

$$\lim_{m \rightarrow \infty} \gamma^m = 2 \int_1^\infty \log(v) v^{-\alpha-1} dv = \frac{2}{\alpha^2} = 2\gamma^2.$$

Step 5: uniform negligibility of the far tail. By Theorem 1.5 (approximation of random elements) it suffices to prove, for every $\eta > 0$,

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\log t}{t} \tilde{T}_n(t) dt > \eta \right) = 0.$$

Fix $\delta \in (0, 1/\gamma)$ and $\varepsilon \in (0, 1/\gamma - \delta)$, and take m large enough that $\log t \leq t^\delta$ for $t \geq m$. Since $\mathbb{E} \tilde{T}_n(t) = T_n(t) = \bar{F}(u_n t) / \bar{F}(u_n) \leq \text{cst } t^{-1/\gamma+\varepsilon}$ for $t \geq 1$ by Potter's bound (Theorem 3.6, with $x = u_n$, $y = 1/t \leq 1$),

$$\mathbb{E} \int_m^\infty \frac{\log t}{t} \tilde{T}_n(t) dt \leq \text{cst} \int_m^\infty t^{-1/\gamma-1+\delta+\varepsilon} dt = O(m^{-1/\gamma+\delta+\varepsilon}),$$

uniformly in n , and this vanishes as $m \rightarrow \infty$. Markov's inequality gives the claim, so $M_n^{(2)} \xrightarrow{\mathbb{P}} 2\gamma^2$.

Step 6: conclusion. Since $\gamma > 0$ we have $M_n^{(1)} \xrightarrow{\mathbb{P}} \gamma > 0$ and $M_n^{(2)} \xrightarrow{\mathbb{P}} 2\gamma^2 > 0$, hence

$$1 - \frac{(M_n^{(1)})^2}{M_n^{(2)}} \xrightarrow{\mathbb{P}} 1 - \frac{\gamma^2}{2\gamma^2} = \frac{1}{2} \neq 0,$$

so the map is continuous at the limit point and

$$\gamma_{n,k}^M = M_n^{(1)} + 1 - \frac{1}{2} \left(1 - \frac{(M_n^{(1)})^2}{M_n^{(2)}} \right)^{-1} \xrightarrow{\mathbb{P}} \gamma + 1 - \frac{1}{2} \cdot 2 = \gamma.$$

Remark 0.6. The structure of the proof is identical to that of Theorem 3.13, with $\log t/t$ in place of $1/t$: consistency on compacts plus a Potter majorant for the far tail. The only genuinely new ingredient is the extra factor $\log t$, which is absorbed by the crude bound $\log t \leq t^\delta$ at the cost of shrinking the admissible ε .

Exercise 10 A log-perturbed tail: $k = o(\log^2 n)$ suffices

Solution. Assume F is continuous on $\mathbb{R}$ and that there are $c > 0$, $\gamma > 0$ and β with

$$\sup_{t \geq 1} t^\beta \left| \bar{F}(t) - c \log(t) t^{-1/\gamma} \right| < \infty. \quad (6)$$

As in Proposition 3.1 we assume $\beta > 1/\gamma$; this is what makes the displayed term the leading one (if $\beta \leq 1/\gamma$ the remainder dominates and $\bar{F}$ need not even be regularly varying with index $1/\gamma$). Write $\bar{F}(t) = c t^{-1/\gamma} \log t + R(t)$ with $|R(t)| \leq C t^{-\beta}$ for $t \geq 1$; note that $\bar{F}$ is regularly varying with index $1/\gamma$ and slowly varying part $\ell(t) \sim c \log t$, the situation of Example 3.4.

Step 1: the threshold. With $u_n = F^{\leftarrow}(1 - k/n)$ we have $\bar{F}(u_n) = k/n$ and $u_n \rightarrow \infty$, so

$$\frac{k}{n} = c u_n^{-1/\gamma} \log u_n (1 + o(1)), \quad \text{hence} \quad \log u_n = \gamma \log(n/k) (1 + o(1)),$$

$$u_n = O\left(\left(\frac{n}{k} \log \frac{n}{k}\right)^\gamma\right).$$

Step 2: the bias of T_n . For $t \geq 1$,

$$\begin{aligned} \bar{F}(u_n t) - t^{-1/\gamma} \bar{F}(u_n) &= c u_n^{-1/\gamma} t^{-1/\gamma} [\log(u_n t) - \log u_n] + R(u_n t) - t^{-1/\gamma} R(u_n) \\ &= c u_n^{-1/\gamma} t^{-1/\gamma} \log t + O(u_n^{-\beta} t^{-1/\gamma}), \end{aligned}$$

where we used $t^{-\beta} \leq t^{-1/\gamma}$ for $t \geq 1$. Dividing by $\bar{F}(u_n) = c u_n^{-1/\gamma} \log u_n (1 + o(1))$,

$$T_n(t) - t^{-1/\gamma} = \frac{t^{-1/\gamma} \log t}{\log u_n} (1 + o(1)) + O\left(\frac{u_n^{1/\gamma-\beta}}{\log u_n}\right) t^{-1/\gamma},$$

uniformly in $t \geq 1$.

Step 3: the integrated bias. Using $\int_1^\infty t^{-1/\gamma-1} \log t \, dt = \gamma^2$ and $\int_1^\infty t^{-1/\gamma-1} dt = \gamma$,

$$\int_1^\infty \frac{T_n(t) - T(t)}{t} dt = \frac{\gamma^2}{\log u_n} (1 + o(1)) + O(u_n^{1/\gamma-\beta}) = \frac{\gamma}{\log(n/k)} (1 + o(1)) + O(u_n^{1/\gamma-\beta}).$$

This recovers the exact bias $\gamma^2 / \log u_n$ of Example 3.4.

Step 4: the rate condition (3.7). Multiplying by $\sqrt{k}$,

$$\sqrt{k} \int_1^\infty \frac{T_n(t) - T(t)}{t} dt = \gamma \frac{\sqrt{k}}{\log(n/k)} (1 + o(1)) + O\left(\sqrt{k} \left(\frac{n}{k} \log \frac{n}{k}\right)^{1-\gamma\beta}\right).$$

Assume now $k = o(\log^2 n)$. Then $\log k = O(\log \log n) = o(\log n)$, so $\log(n/k) = \log n (1 + o(1))$ and the first term satisfies

$$\gamma \frac{\sqrt{k}}{\log(n/k)} \sim \gamma \frac{\sqrt{k}}{\log n} = \gamma \frac{o(\log n)}{\log n} \rightarrow 0.$$

The second term vanishes as well: $\gamma\beta > 1$, so $(n/k)^{1-\gamma\beta}$ decays polynomially in n while $\sqrt{k} \leq \log n$ and the logarithmic factor is negligible. Hence (3.7) holds, i.e. $k = o(\log^2 n)$ is a suitable choice.

Remark 0.7. The conclusion is sharp in order: Step 4 shows (3.7) is *equivalent* to $\sqrt{k} / \log(n/k) \rightarrow 0$, i.e. $k = o(\log^2(n/k))$, so no polynomial $k = n^\rho$ is admissible, in stark contrast with Proposition 3.1 (where the pure power tail allows $k = o(n^\rho)$ with $\rho = 2(\beta\gamma - 1)/(2\beta\gamma - 1)$). This is exactly the pathology of Example 3.4: with $n = 10^5$ one needs $k \ll \log^2 n \approx 25$ effective order statistics, so the asymptotic normality of the Hill estimator is a practically empty statement for such tails.

Exercise 11 Proof of Lemma 3.2 (kernel tail expectation)

Solution. Claim. Under conditions 3.1.1–3.1.3, for any $t, q > 0$,

$$\frac{\mathbb{E}[\mathbf{1}(Y_1 > u_n^x t) K^q\left(\frac{x-X_1}{h_n}\right)]}{h_n \bar{F}^x(u_n^x)} \rightarrow t^{-1/\gamma(x)} g(x) \int K^q(u) du.$$

Step 1: conditioning on the covariate and rescaling. Conditioning on $X_1 = z$ and substituting $z = x - h_n u$ (so $dz = h_n du$; K is supported on $[-1, 1]$, so u ranges over $[-1, 1]$ and z over $[x - h_n, x + h_n] \subseteq U$ for n large),

$$\mathbb{E}\left[\mathbf{1}(Y_1 > u_n^x t) K^q\left(\frac{x-X_1}{h_n}\right)\right] = \int K^q\left(\frac{x-z}{h_n}\right) \bar{F}^z(u_n^x t) g(z) dz$$

$$= h_n \int_{-1}^1 K^q(u) \bar{F}^{x-h_n u}(u_n^x t) g(x - h_n u) du.$$

Hence the quantity of interest equals

$$\int_{-1}^1 K^q(u) \underbrace{\frac{\bar{F}^{x-h_n u}(u_n^x t)}{\bar{F}^x(u_n^x t)}}_{=: A_n(u)} \cdot \underbrace{\frac{\bar{F}^x(u_n^x t)}{\bar{F}^x(u_n^x)}}_{=: T_n^x(t)} \cdot g(x - h_n u) du. \quad (7)$$

Step 2: the three factors.

- $T_n^x(t) \rightarrow t^{-1/\gamma(x)}$: by (3.14), $\bar{F}^x(y) = L_x(y)y^{-1/\gamma(x)}$ with L_x slowly varying, so $T_n^x(t) = t^{-1/\gamma(x)} L_x(u_n^x t) / L_x(u_n^x) \rightarrow t^{-1/\gamma(x)}$ as $u_n^x \rightarrow \infty$.
- $A_n(u) \rightarrow 1$ uniformly in $u \in [-1, 1]$: by (3.14) applied at $x - h_n u$ and at x , with $y := u_n^x t$,

$$A_n(u) = \frac{L_{x-h_n u}(y)}{L_x(y)} \cdot y^{-\left(\frac{1}{\gamma(x-h_n u)} - \frac{1}{\gamma(x)}\right)}.$$

Condition 3.1.1 (c_γ -Lipschitz $1/\gamma$) gives

$$\left| \log y^{-(1/\gamma(x-h_n u) - 1/\gamma(x))} \right| \leq c_\gamma h_n |u| \log y \leq c_\gamma h_n (\log u_n^x + |\log t|) \rightarrow 0$$

by condition 3.1.2 ($h_n \log u_n^x \rightarrow 0$) and $h_n \rightarrow 0$. Condition 3.1.3 gives, for $y \geq y_0$,

$$\left| \log \frac{L_{x-h_n u}(y)}{L_x(y)} \right| \leq c_L h_n |u| \log y \rightarrow 0$$

by the same argument. Both bounds are uniform in $u \in [-1, 1]$.

- $g(x - h_n u) \rightarrow g(x)$ uniformly in $u \in [-1, 1]$, since g is (twice) continuously differentiable, hence locally Lipschitz.

Step 3: dominated convergence. By Step 2 the integrand of (7) converges pointwise to $t^{-1/\gamma(x)} g(x) K^q(u)$ and, for n large, is bounded by $\text{cst } K^q(u)$, which is integrable on $[-1, 1]$. Therefore

$$\frac{\mathbb{E}[\mathbf{1}(Y_1 > u_n^x t) K^q(\frac{x-X_1}{h_n})]}{h_n \bar{F}^x(u_n^x)} \rightarrow t^{-1/\gamma(x)} g(x) \int_{-1}^1 K^q(u) du,$$

as claimed. $\square$

Two special cases are the ones used in the chapter: $q = 1$ gives $\mathbb{E} \tilde{V}_n^x(t) \rightarrow t^{-1/\gamma(x)} g(x)$ (Lemma 3.1, with $\int K = 1$), and $q = 2$ supplies the second-moment term in $\text{Var}(\tilde{V}_n^x)$ and in the covariance computation of Lemma 3.3; the factor $\int K^2(u) du$ appearing in the asymptotic variance of the conditional Hill estimator originates precisely here.

Exercise 12 Variance of the conditional Hill limit

Solution. Abbreviate $\alpha := 1/\gamma(x)$, $g := g(x)$ and $\kappa := \int K^2(u) du$. The limit in Theorem 3.26 is

$$\xi := \frac{1}{g} \int_1^\infty \frac{\mathbb{V}^x(s)}{s} ds - \frac{\gamma(x)}{g} \mathbb{V}^x(1) = \int_1^\infty \frac{\mathbb{T}^x(s)}{s} ds - \gamma(x) \mathbb{T}^x(1),$$

where $\mathbb{T}^x = \mathbb{V}^x/g$ is the centred Gaussian process with covariance (3.19),

$$\Gamma(s, t) = \text{Cov}(\mathbb{T}^x(s), \mathbb{T}^x(t)) = \frac{(s \vee t)^{-\alpha}}{g} \kappa.$$

(Absolute convergence of the integral in L^1 , hence the validity of Fubini below, follows as in Exercise 8 from $\mathbb{E}|\mathbb{T}^x(s)| \leq \text{cst} s^{-\alpha/2}$.) Expanding the variance of the difference,

$$\text{Var}(\xi) = \underbrace{\int_1^\infty \int_1^\infty \frac{\Gamma(s, t)}{st} ds dt}_{=:A} - 2\gamma(x) \underbrace{\int_1^\infty \frac{\Gamma(s, 1)}{s} ds}_{=:B} + \gamma(x)^2 \Gamma(1, 1).$$

Now, using symmetry and $\int_1^t s^{-1} ds = \log t$,

$$A = \frac{\kappa}{g} \cdot 2 \int_1^\infty t^{-\alpha-1} \int_1^t \frac{ds}{s} dt = \frac{2\kappa}{g} \int_1^\infty t^{-\alpha-1} \log t dt = \frac{2\kappa}{g} \cdot \frac{1}{\alpha^2} = \frac{2\kappa\gamma(x)^2}{g},$$

$$B = \frac{\kappa}{g} \int_1^\infty s^{-\alpha-1} ds = \frac{\kappa}{g\alpha} = \frac{\kappa\gamma(x)}{g}, \quad \Gamma(1, 1) = \frac{\kappa}{g}.$$

Therefore

$$\text{Var}(\xi) = \frac{2\kappa\gamma(x)^2}{g} - 2\gamma(x) \frac{\kappa\gamma(x)}{g} + \gamma(x)^2 \frac{\kappa}{g} = \frac{\gamma(x)^2}{g(x)} \int K^2(u) du,$$

which is the asymptotic variance in Theorem 3.26. As ξ is a centred Gaussian variable, $\sqrt{k_n h_n}(\gamma_{n,k}(x) - \gamma(x)) \xrightarrow{d} N(0, \gamma(x)^2 g(x)^{-1} \int K^2)$.

Remark 0.8. Exercise 8 is the special case $g \equiv 1$, $\kappa = 1$: the three terms are $2\gamma^2 - 2\gamma^2 + \gamma^2 = \gamma^2$. The conditional variance differs from the unconditional one by the factor $g(x)^{-1} \int K^2$, the usual kernel-smoothing inflation: it is large where covariate observations are sparse ($g(x)$ small), and $\int K^2$ is minimised by the Epanechnikov kernel. Note also that the rate is $\sqrt{k_n h_n}$, not $\sqrt{k_n}$: the effective sample size is $nh_n \bar{F}^x(u_n^x) = k_n h_n$.

Exercise 13 A uniformity detail in the proof of Theorem 3.26

Solution. We must show that

$$\lim_{n \rightarrow \infty} \sup_{u \in [x-h_n, x+h_n]} \left| \frac{1}{\bar{F}^u(u_n^x)} \mathbb{E} \left[\log_+^2 \left(\frac{Y_1}{u_n^x m} \right) \middle| X_0 = u \right] - \frac{1}{\bar{F}^x(u_n^x)} \mathbb{E} \left[\log_+^2 \left(\frac{Y_1}{u_n^x m} \right) \middle| X_0 = x \right] \right| = 0. \quad (8)$$

Step 1: rewrite as a tail integral. For a non-negative variable V , $\mathbb{E}[\log_+^2 V] = \int_0^\infty \mathbb{P}(\log_+^2 V > \lambda) d\lambda = 2 \int_0^\infty w \mathbb{P}(V > e^w) dw$, so with $V = Y_1/(u_n^x m)$ and $t = e^w$,

$$A_n(u) := \frac{1}{\bar{F}^u(u_n^x)} \mathbb{E} \left[\log_+^2 \left(\frac{Y_1}{u_n^x m} \right) \middle| X_0 = u \right] = 2 \int_1^\infty \frac{\log t}{t} R_n^u(t) dt, \quad R_n^u(t) := \frac{\bar{F}^u(mt u_n^x)}{\bar{F}^u(u_n^x)}.$$

For $u = x$ this converges, by conditional regular variation and dominated convergence (Potter), to $\int_m^\infty \log^2(t) \gamma(x)^{-1} t^{-1/\gamma(x)-1} dt$ up to the change of variable used in the chapter; only the *difference* matters here.

Step 2: compare R_n^u and R_n^x . By (3.14),

$$\frac{R_n^u(t)}{R_n^x(t)} = \exp(z_n(t, u)), \quad z_n(t, u) = \log \frac{L_u(y)}{L_x(y)} - \log \frac{L_u(u_n^x)}{L_x(u_n^x)} - \left(\frac{1}{\gamma(u)} - \frac{1}{\gamma(x)} \right) \log \frac{y}{u_n^x},$$

with $y := mt u_n^x$. Conditions 3.1.1 and 3.1.3, together with $|u - x| \leq h_n$, give

$$|z_n(t, u)| \leq c_L h_n (\log y + \log u_n^x) + c_\gamma h_n \log(mt) \leq \text{cst } h_n (\log(mt) + \log u_n^x) =: \zeta_n(t),$$

uniformly in $u \in [x - h_n, x + h_n]$.

Step 3: linearise the exponential. With $|e^z - 1| \leq |z| + \frac{1}{2}z^2 e^{|z|}$,

$$|R_n^u(t) - R_n^x(t)| = R_n^x(t) |e^{z_n} - 1| \leq R_n^x(t) \zeta_n(t) \left(1 + \frac{1}{2}\zeta_n(t) e^{\zeta_n(t)}\right), \quad e^{\zeta_n(t)} = \text{cst} (mtu_n^x)^{\text{cst} h_n}.$$

By Potter's bound in the conditional setting, $R_n^x(t) \leq \text{cst} (mt)^{-1/\gamma(x)+\varepsilon}$ for $\varepsilon \in (0, 1/\gamma(x))$. Choose $\varepsilon' \in (0, 1/\gamma(x) - \varepsilon)$ and n' with $\text{cst} h_n \leq \varepsilon'$ for $n \geq n'$; also $(u_n^x)^{\text{cst} h_n} = \exp(\text{cst} h_n \log u_n^x) = O(1)$ by condition 3.1.2. Hence, for $n \geq n'$ and $t \geq 1$,

$$|R_n^u(t) - R_n^x(t)| \leq \text{cst} h_n (\log(mt) + \log u_n^x) \left(1 + h_n (\log(mt) + \log u_n^x)\right) (mt)^{-1/\gamma(x)+\varepsilon+\varepsilon'}.$$

Step 4: integrate. Since $-1/\gamma(x) + \varepsilon + \varepsilon' < 0$, all the integrals $\int_1^\infty t^{-1/\gamma(x)+\varepsilon+\varepsilon'-1} \log^j(t) dt$, $j = 0, 1, 2, 3$, are finite constants. Therefore

$$\sup_{|u-x| \leq h_n} |A_n(u) - A_n(x)| \leq 2 \int_1^\infty \frac{\log t}{t} |R_n^u(t) - R_n^x(t)| dt = O(h_n \log u_n^x) + O(h_n) + O(h_n^2 \log^2 u_n^x),$$

and each term tends to 0 by condition 3.1.2 ($h_n \log u_n^x \rightarrow 0$) and $h_n \rightarrow 0$. This proves (8). $\square$

The role of the limit in Theorem 3.26 is to replace, inside the variance bound for $\int_m^\infty \tilde{V}_n^x(s) s^{-1} ds$, the local averages over the kernel window $[x - h_n, x + h_n]$ by the value at the centre x ; the display above shows that the error committed is $O(h_n \log u_n^x) = o(1)$, which is precisely the content of condition 3.1.2.

Exercise 14 Second-order condition (3.20) implies Condition 3.2

Throughout this exercise and the next, $\alpha := 1/\gamma(x)$, $\kappa := \int K^2$, and we assume Condition 3.1, F^x continuous, and

$$\sup_{t \geq 1} t^{\beta_x} |\bar{F}^x(t) - c_x t^{-1/\gamma(x)}| < \infty, \quad c_x > 0, \quad \beta_x > 1/\gamma(x) > 0. \quad (3.20)$$

The goal is: if

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n^3} \log(n/k_n) = \lim_{n \rightarrow \infty} \sqrt{h_n} k_n^{\gamma(x)\beta_x - \frac{1}{2}} n^{1-\gamma(x)\beta_x} = 0, \quad (3.21)$$

then Condition 3.2 holds.

(a) Order of u_n^x and the three-term decomposition

Solution. Order of the threshold. Write $\bar{F}^x(t) = c_x t^{-\alpha} + R(t)$ with $|R(t)| \leq Ct^{-\beta_x}$. Since $\bar{F}^x(u_n^x) = k_n/n$ and $u_n^x \rightarrow \infty$,

$$\frac{k_n}{n} = c_x (u_n^x)^{-\alpha} \left(1 + O((u_n^x)^{-\beta_x+\alpha})\right) = c_x (u_n^x)^{-\alpha} (1 + o(1)),$$

because $\beta_x > \alpha$. Hence $(u_n^x)^\alpha = c_x (n/k_n) (1 + o(1))$, i.e.

$$u_n^x = O((n/k_n)^{\gamma(x)}), \quad \log u_n^x = \gamma(x) \log(n/k_n) (1 + o(1)).$$

The decomposition. By the computation in Lemma 3.1 / Exercise 11 (with $q = 1$, $\int K = 1$),

$$\mathbb{E}[\tilde{V}_n^x(t)] = \int K(u) \frac{\bar{F}^{x-uh_n}(tu_n^x)}{\bar{F}^x(u_n^x)} g(x - h_n u) du.$$

Adding and subtracting $\bar{F}^x(tu_n^x)/\bar{F}^x(u_n^x)$ inside the integral, and using $\int K(u)du = 1$ to write $t^{-\alpha}g(x) = \int K(u)t^{-\alpha}g(x)du$, we obtain

$$\mathbb{E}[\tilde{V}_n^x(t)] - t^{-1/\gamma(x)}g(x) = \int K(u)t^{-1/\gamma(x)}(g(x - h_n u) - g(x))du \quad (\text{I})$$

$$+ \int K(u) \left(\frac{\bar{F}^{x-uh_n}(tu_n^x)}{\bar{F}^x(u_n^x)} - \frac{\bar{F}^x(tu_n^x)}{\bar{F}^x(u_n^x)} \right) g(x - h_n u) du \quad (\text{II})$$

$$+ \int K(u) \left(\frac{\bar{F}^x(tu_n^x)}{\bar{F}^x(u_n^x)} - t^{-1/\gamma(x)} \right) g(x - h_n u) du, \quad (\text{III})$$

which is the claimed decomposition: (I) is a pure smoothing (kernel) bias, (II) is the local variation of the conditional law across the bandwidth window, and (III) is the second-order (extreme-value) bias at the centre x .

(b) The smoothing term (I)

Solution. g is twice continuously differentiable, hence Lipschitz on the compact $[x - 1, x + 1]$, with constant $c_g := \sup_{|v-x|\leq 1} |g'(v)|$. As K is supported on $[-1, 1]$,

$$\left| \int K(u)t^{-\alpha}(g(x - h_n u) - g(x))du \right| \leq t^{-\alpha}c_g h_n \int_{-1}^1 |u| |K(u)| du = O(t^{-1/\gamma(x)}h_n).$$

Remark 0.9. If in addition K is symmetric, $\int uK(u)du = 0$, and a second-order Taylor expansion gives the sharper $O(t^{-\alpha}h_n^2)$; this is why g is assumed *twice* differentiable. The rate conditions below only use the cruder $O(h_n)$.

(c) The local-variation term (II)

Solution. Fix $u \in [-1, 1]$ and set $y := tu_n^x$. By (3.14),

$$\frac{\bar{F}^{x-uh_n}(y)}{\bar{F}^x(y)} = e^{z_n}, \quad z_n = \log \frac{L_{x-uh_n}(y)}{L_x(y)} - \left(\frac{1}{\gamma(x-uh_n)} - \frac{1}{\gamma(x)} \right) \log y,$$

so that, by conditions 3.1.1 and 3.1.3,

$$|z_n| \leq (c_\gamma + c_L) |u| h_n \log(tu_n^x) =: \zeta.$$

Using $|e^z - 1| \leq |z| + \frac{1}{2}z^2e^{|z|}$ and Potter's bound in the conditional setting, $\bar{F}^x(tu_n^x)/\bar{F}^x(u_n^x) \leq \text{cst } t^{-\alpha+\varepsilon}$ for $\varepsilon \in (0, 1/\gamma(x))$,

$$\begin{aligned} \left| \frac{\bar{F}^{x-uh_n}(tu_n^x)}{\bar{F}^x(u_n^x)} - \frac{\bar{F}^x(tu_n^x)}{\bar{F}^x(u_n^x)} \right| &= \frac{\bar{F}^x(tu_n^x)}{\bar{F}^x(u_n^x)} |e^{z_n} - 1| \\ &\leq \text{cst } t^{-1/\gamma(x)+\varepsilon} \log(tu_n^x) |u| h_n \left(1 + \frac{1}{2} \log(tu_n^x) |u| h_n (tu_n^x)^{(c_\gamma+c_L)|u|h_n} \right), \end{aligned}$$

which is the bound in the statement (there $e^\zeta = (tu_n^x)^{(c_\gamma+c_L)|u|h_n}$). Since g is bounded on the window and $\int |K| < \infty$, the same bound (with a new constant) holds for the whole term (II).

Leading order. The factor in brackets is $1+o(1)$: indeed $h_n \log(tu_n^x) \rightarrow 0$ by 3.1.2 and $(u_n^x)^{(c_\gamma+c_L)h_n} = \exp((c_\gamma + c_L)h_n \log u_n^x) \rightarrow 1$. Using $\log u_n^x = \gamma(x) \log(n/k_n)(1 + o(1))$ from (a), the leading term is

$$O\left(t^{-1/\gamma(x)+\varepsilon} \log(tn/k_n) h_n\right), \quad \varepsilon \in (0, 1/\gamma(x)).$$

(d) The second-order term (III)

Solution. For $t \geq 1$, using (3.20) twice and $t^{-\beta_x} \leq t^{-\alpha}$ (valid because $\beta_x > \alpha$ and $t \geq 1$),

$$|\bar{F}^x(tu_n^x) - t^{-\alpha}\bar{F}^x(u_n^x)| \leq |\bar{F}^x(tu_n^x) - c_x(tu_n^x)^{-\alpha}| + t^{-\alpha}|c_x(u_n^x)^{-\alpha} - \bar{F}^x(u_n^x)| \leq 2C(u_n^x)^{-\beta_x}t^{-\alpha}.$$

Dividing by $\bar{F}^x(u_n^x) = k_n/n$ and inserting $u_n^x \asymp (n/k_n)^{\gamma(x)}$ from (a),

$$\left| \frac{\bar{F}^x(tu_n^x)}{\bar{F}^x(u_n^x)} - t^{-1/\gamma(x)} \right| \leq \text{cst} \frac{n}{k_n} (u_n^x)^{-\beta_x} t^{-\alpha} = O\left(k_n^{\beta_x \gamma(x)-1} n^{1-\beta_x \gamma(x)} t^{-1/\gamma(x)}\right),$$

uniformly in $t \geq 1$; the same bound holds for (III) since g is bounded and $\int |K| < \infty$. This is the exact analogue of the computation in Proposition 3.1.

(e) Condition 3.2.1

Solution. Let $s_0 \in (0, 1)$. For $t \geq s_0$ the maps $t \mapsto t^{-\alpha}$ and $t \mapsto t^{-\alpha+\varepsilon} \log t$ are bounded (the latter because $-\alpha + \varepsilon < 0$), so taking suprema in (b), (c), (d):

$$B_n^x(s_0) = \sup_{t \geq s_0} |\mathbb{E} \tilde{V}_n^x(t) - t^{-1/\gamma(x)} g(x)| = O(h_n) + O(h_n \log(n/k_n)) + O(k_n^{\beta_x \gamma(x)-1} n^{1-\beta_x \gamma(x)}).$$

For the middle term one uses, as in Theorem 3.21, the choice $\varepsilon' \in (0, 1/\gamma(x) - \varepsilon)$ and n' such that $(c_\gamma + c_L)h_n \leq \varepsilon'$ for $n \geq n'$, so that $t^{-\alpha+\varepsilon+\varepsilon'}$ still decays and $(u_n^x)^{(c_\gamma+c_L)h_n} = O(1)$; all three O -terms are then uniform in $t \geq s_0$. Multiplying by $\sqrt{k_n h_n}$,

$$\sqrt{k_n h_n} B_n^x(s_0) = O\left(\underbrace{\sqrt{k_n h_n^3} \log(n/k_n)}_{\rightarrow 0 \text{ by (3.21)}}\right) + O\left(\underbrace{\sqrt{h_n} k_n^{\beta_x \gamma(x)-\frac{1}{2}} n^{1-\beta_x \gamma(x)}}_{\rightarrow 0 \text{ by (3.21)}}\right) \rightarrow 0,$$

where we absorbed $\sqrt{k_n h_n^3} = \sqrt{k_n h_n} h_n$ and noted that the first O -term dominates $\sqrt{k_n h_n} h_n$ since $\log(n/k_n) \rightarrow \infty$. Hence condition 3.2.1 holds.

(f) The integrated decomposition

Solution. Integrating the bounds of (b), (c), (d) against dt/t over $[1, \infty)$ gives immediately

$$\begin{aligned} & \int_1^\infty \frac{|\mathbb{E}[\tilde{V}_n^x(t)] - t^{-1/\gamma(x)} g(x)|}{t} dt \\ & \leq O(h_n) \int_1^\infty t^{-1/\gamma(x)-1} dt + O(k_n^{\beta_x \gamma(x)-1} n^{1-\beta_x \gamma(x)}) \int_1^\infty t^{-1/\gamma(x)-1} dt \\ & \quad + O(h_n) \int_1^\infty t^{-1/\gamma(x)+\varepsilon-1} \log(tu_n^x) \left(1 + \frac{h_n}{2} \log(tu_n^x) (tu_n^x)^{(c_\gamma+c_L)h_n}\right) dt, \end{aligned}$$

which is the displayed inequality. The first two integrals equal $\gamma(x)$, so the first two terms are $O(h_n)$ and $O(k_n^{\beta_x \gamma(x)-1} n^{1-\beta_x \gamma(x)})$.

(g) Condition 3.2.2

Solution. It remains to bound the third integral,

$$J_n := O(h_n) \int_1^\infty t^{-1/\gamma(x)+\varepsilon-1} \log(tu_n^x) \left(1 + \frac{1}{2} \log(tu_n^x) h_n (tu_n^x)^{(c_\gamma+c_L)h_n}\right) dt.$$

Preliminaries (as in the hint). By condition 3.1.2, $(u_n^x)^{h_n} = \exp(h_n \log u_n^x) \rightarrow 1$, so $(u_n^x)^{h_n} = O(1)$; and there is n' such that $(c_\gamma + c_L)h_n \leq \varepsilon' \in (0, 1/\gamma(x) - \varepsilon)$ for all $n \geq n'$. Fix $n \geq n'$ and split $\log(tu_n^x) = \log t + \log u_n^x$. Then, with $\alpha - \varepsilon - \varepsilon' > 0$, all the integrals

$$I_j := \int_1^\infty t^{-\alpha+\varepsilon+\varepsilon'-1} \log^j t \, dt < \infty, \quad j = 0, 1, 2,$$

are finite constants not depending on n , and

$$J_n \leq \text{cst } h_n \left[(I_1 + I_0 \log u_n^x) + h_n O(1) (I_2 + 2I_1 \log u_n^x + I_0 \log^2 u_n^x) \right] = O(h_n \log u_n^x) + O(h_n^2 \log^2 u_n^x).$$

Both terms are bounded — indeed they tend to 0 — because $h_n \log u_n^x \rightarrow 0$ by condition 3.1.2. Using $\log u_n^x = \gamma(x) \log(n/k_n)(1 + o(1))$ from (a), $J_n = O(h_n \log(n/k_n))$.

Combining with (f),

$$\begin{aligned} & \sqrt{k_n h_n} \int_1^\infty \frac{|\mathbb{E}[\tilde{V}_n^x(t)] - t^{-1/\gamma(x)} g(x)|}{t} dt \\ &= O\left(\sqrt{k_n h_n^3} \log(n/k_n)\right) + O\left(\sqrt{h_n} k_n^{\beta_x \gamma(x) - \frac{1}{2}} n^{1 - \beta_x \gamma(x)}\right) \rightarrow 0 \end{aligned}$$

by (3.21). Hence condition 3.2.2 holds, and together with (e), Condition 3.2 holds in full. $\square$

Exercise 15 The log-perturbed second-order condition

Solution. Assume now, instead of (3.20),

$$\sup_{t \geq 1} t^{\beta_x} |\bar{F}^x(t) - c_x \log(t) t^{-1/\gamma(x)}| < \infty, \quad c_x > 0, \quad \beta_x > 1/\gamma(x) > 0, \quad (3.22)$$

and

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n^3} \log(n/k_n) = \lim_{n \rightarrow \infty} \frac{\sqrt{k_n h_n}}{\log(n/k_n)} = 0. \quad (3.23)$$

The decomposition (I)+(II)+(III) of Exercise 14(a) and the bounds for (I) and (II) in (b)–(c) are unchanged, since they never used the form of the second-order term. Only (III) and the threshold asymptotics change.

Step 1: threshold. Writing $\bar{F}^x(t) = c_x \log(t) t^{-\alpha} + R(t)$, $|R(t)| \leq C t^{-\beta_x}$, and $\bar{F}^x(u_n^x) = k_n/n$,

$$\frac{k_n}{n} = c_x \log(u_n^x) (u_n^x)^{-\alpha} (1 + o(1)),$$

$$\log u_n^x = \gamma(x) \log(n/k_n) (1 + o(1)), \quad u_n^x = O\left(\left(\frac{n}{k_n} \log \frac{n}{k_n}\right)^{\gamma(x)}\right).$$

Step 2: the new term (III). Exactly as in Exercise 10, for $t \geq 1$,

$$\bar{F}^x(tu_n^x) - t^{-\alpha} \bar{F}^x(u_n^x) = c_x (u_n^x)^{-\alpha} t^{-\alpha} \log t + O((u_n^x)^{-\beta_x} t^{-\alpha}),$$

so, dividing by $\bar{F}^x(u_n^x) = c_x (u_n^x)^{-\alpha} \log u_n^x (1 + o(1))$,

$$\left| \frac{\bar{F}^x(tu_n^x)}{\bar{F}^x(u_n^x)} - t^{-1/\gamma(x)} \right| \leq \frac{t^{-\alpha} \log t}{\log u_n^x} (1 + o(1)) + O\left(\left(\frac{n}{k_n} (u_n^x)^{-\beta_x}\right) t^{-\alpha}\right), \quad (9)$$

uniformly in $t \geq 1$. The dominant part is now of order $1/\log(n/k_n)$ instead of $(n/k_n)^{1-\beta_x \gamma(x)}$: the bias decays only logarithmically.

Step 3: condition 3.2.1. Taking suprema over $t \geq s_0$ in (I), (II) and (9),

$$B_n^x(s_0) = O(h_n) + O(h_n \log(n/k_n)) + O\left(\frac{1}{\log(n/k_n)}\right) + O\left(\left(\frac{n}{k_n}\right)^{1-\beta_x \gamma(x)} \log^{-\beta_x \gamma(x)} \frac{n}{k_n}\right).$$

Multiplying by $\sqrt{k_n h_n}$: the first two terms give $O(\sqrt{k_n h_n^3} \log(n/k_n)) \rightarrow 0$ by (3.23), the third gives $\sqrt{k_n h_n} / \log(n/k_n) \rightarrow 0$ by (3.23). The fourth is automatic: (3.23) forces $\sqrt{k_n h_n} = o(\log(n/k_n))$, hence

$$\sqrt{k_n h_n} \left(\frac{n}{k_n} \right)^{1-\beta_x \gamma(x)} = o \left(\log \frac{n}{k_n} \left(\frac{n}{k_n} \right)^{1-\beta_x \gamma(x)} \right) \rightarrow 0,$$

since $\beta_x \gamma(x) > 1$ and $n/k_n \rightarrow \infty$. So condition 3.2.1 holds.

Step 4: condition 3.2.2. Integrating (9) against dt/t and using $\int_1^\infty t^{-\alpha-1} \log t \, dt = \gamma(x)^2$,

$$\int_1^\infty \frac{|(\text{III})|}{t} dt = \frac{\gamma(x)^2}{\log u_n^x} (1 + o(1)) + O \left(\left(\frac{n}{k_n} \right)^{1-\beta_x \gamma(x)} \right) = O \left(\frac{1}{\log(n/k_n)} \right) + O \left(\left(\frac{n}{k_n} \right)^{1-\beta_x \gamma(x)} \right),$$

while the contributions of (I) and (II) are $O(h_n)$ and $O(h_n \log(n/k_n))$ exactly as in Exercise 14(f)–(g). Multiplying by $\sqrt{k_n h_n}$ and using (3.23) together with the argument of Step 3 gives condition 3.2.2. Hence Condition 3.2 holds. $\square$

Remark 0.10. The contrast with Exercise 14 is the conditional analogue of Example 3.4 versus Proposition 3.1. Under (3.20) the bias is polynomially small and (3.21) permits polynomial k_n ; under (3.22) the bias is only $O(1/\log(n/k_n))$, so (3.23) forces the *effective* sample size $k_n h_n$ to grow more slowly than $\log^2(n/k_n)$, and the conditional Hill estimator converges at rate at best $\log(n/k_n)$.

Exercise 16 Admissible $(k_n), (h_n)$ under (3.20)

Solution. Assume conditions 3.1.1, 3.1.3 and (3.20); we must exhibit $(k_n), (h_n)$ for which all hypotheses of Theorems 3.21 and 3.26 hold, namely

$$\begin{aligned} \text{(i)} \quad k_n \rightarrow \infty, \quad k_n/n \rightarrow 0, \quad h_n \rightarrow 0; \quad & \text{(ii)} \quad 3.1.2 : h_n \log u_n^x \rightarrow 0; \\ \text{(iii)} \quad 3.1.4 : k_n h_n \rightarrow \infty; \quad & \text{(iv)} \quad (3.21). \end{aligned}$$

(Condition 3.2 then follows from Exercise 14.) Write $\rho := \gamma(x)\beta_x > 1$ and recall from Exercise 14(a) that $\log u_n^x = \gamma(x) \log(n/k_n)(1 + o(1))$.

A polynomial choice. Take

$$k_n = \lfloor n^a \rfloor, \quad h_n = n^{-b}, \quad 0 < b < a < 1.$$

Then $\log(n/k_n) = (1-a) \log n$ and we check the four groups of conditions:

- (i) holds since $0 < a < 1$ and $b > 0$.
- (ii): $h_n \log u_n^x \asymp n^{-b} \log n \rightarrow 0$ for any $b > 0$.
- (iii): $k_n h_n \asymp n^{a-b} \rightarrow \infty \iff a > b$.
- (iv), first limit: $\sqrt{k_n h_n^3} \log(n/k_n) \asymp n^{(a-3b)/2} \log n \rightarrow 0 \iff a < 3b$.
- (iv), second limit: $\sqrt{h_n} k_n^{\rho-1/2} n^{1-\rho} \asymp n^{-b/2+a(\rho-1/2)+1-\rho} \rightarrow 0 \iff a(\rho - \frac{1}{2}) < \rho - 1 + \frac{b}{2}$.

So it suffices to find $0 < b < a < 3b$, $a < 1$, with $a(\rho - \frac{1}{2}) < \rho - 1 + \frac{b}{2}$. Take $a = 2b$ (which satisfies $b < a < 3b$). The last requirement becomes

$$2b \left(\rho - \frac{1}{2} \right) - \frac{b}{2} < \rho - 1 \iff b < \frac{\rho - 1}{2\rho - \frac{3}{2}},$$

and the right-hand side is strictly positive because $\rho > 1$. Hence any

$$b \in \left(0, \min \left\{ \frac{1}{2}, \frac{\rho - 1}{2\rho - 3/2} \right\} \right), \quad a = 2b, \quad k_n = \lfloor n^{2b} \rfloor, \quad h_n = n^{-b},$$

satisfies every condition for consistency (Theorems 3.20–3.21) and asymptotic normality (Theorem 3.26). With these sequences the effective sample size is $k_n h_n \asymp n^b$ and the rate of convergence of $\gamma_{n,k}(x)$ is $n^{-b/2}$.

Remark 0.11. The bandwidth is undersmoothed on purpose: the first limit in (3.21) requires $h_n^3 k_n \log^2(n/k_n) \rightarrow 0$, which caps h_n , while 3.1.4 requires $k_n h_n \rightarrow \infty$, which floors it; the pair $(a, b) = (2b, b)$ sits comfortably between. Note also that the theoretical choice $h_n = \sqrt{\log(k_n)/n}$ mentioned in Remark 3.1 is of the same nature.

Exercise 17 Admissible $(k_n), (h_n)$ under (3.22)

Solution. Assume conditions 3.1.1, 3.1.3 and (3.22). By Exercise 15 it suffices to find $(k_n), (h_n)$ with (i) $k_n \rightarrow \infty$, $k_n/n \rightarrow 0$, $h_n \rightarrow 0$; (ii) $h_n \log u_n^x \rightarrow 0$; (iii) $k_n h_n \rightarrow \infty$; and (iv) (3.23).

No polynomial choice can work. If $k_n \asymp n^a$ and $h_n \asymp n^{-b}$ then (iii) forces $a > b$, whereas the second limit in (3.23), $\sqrt{k_n h_n} / \log(n/k_n) \asymp n^{(a-b)/2} / \log n \rightarrow 0$, forces $a \leq b$. So the effective sample size $k_n h_n$ must grow slower than any power of n — it may grow only logarithmically.

A valid choice. Take, for any fixed $b \in (0, 1)$,

$$h_n = n^{-b}, \quad k_n = \lceil n^b \log n \rceil, \quad \text{so that} \quad k_n h_n \asymp \log n.$$

Then $\log(n/k_n) = (1 - b) \log n - \log \log n \sim (1 - b) \log n$ and:

- (i): $k_n \rightarrow \infty$, $k_n/n \asymp n^{b-1} \log n \rightarrow 0$, $h_n \rightarrow 0$. ✓
- (ii): $h_n \log u_n^x \asymp n^{-b} \log n \rightarrow 0$ (using $\log u_n^x \sim \gamma(x) \log(n/k_n)$ from Exercise 15, Step 1). ✓
- (iii): $k_n h_n \asymp \log n \rightarrow \infty$. ✓
- (iv), first limit: $\sqrt{k_n h_n^3} \log(n/k_n) = \sqrt{k_n h_n} \cdot h_n \cdot \log(n/k_n) \asymp \sqrt{\log n} n^{-b} \log n \rightarrow 0$. ✓
- (iv), second limit: $\frac{\sqrt{k_n h_n}}{\log(n/k_n)} \asymp \frac{\sqrt{\log n}}{(1 - b) \log n} = \frac{1}{(1 - b) \sqrt{\log n}} \rightarrow 0$. ✓

All the conditions for consistency and for asymptotic normality are therefore satisfied. The resulting rate of convergence is

$$\sqrt{k_n h_n} \asymp \sqrt{\log n},$$

i.e. $\gamma_{n,k}(x) - \gamma(x) = O_{\mathbb{P}}(1/\sqrt{\log n})$.

Remark 0.12. This is the conditional counterpart of the “big bias” pathology of Example 3.4: a slowly varying factor $\log t$ in the conditional tail destroys any polynomial rate, and the central limit theorem, although formally valid, describes a statistic whose effective sample size grows like $\log n$. For $n = 10^6$ this is $k_n h_n \approx 14$ effective observations, so in practice one should regard the normal approximation as indicative only, and read the conditional Hill plot for stability rather than trusting the confidence bands.

Appendix: small corrections to the chapter text

While working through the exercises the following minor points surfaced; none affects any conclusion.

1. *Theorem 3.5, second bound.* The argument “ $z^\varepsilon \ell(z)$ is strictly positive and diverging, consequently $\inf_{x \geq \varepsilon} x^\varepsilon \ell(x) > 0$ ” needs the monotonicity of $\bar{F}$ to handle the compact part; see Exercise 6(a), item (3). Also the bound cannot hold near the origin, which is why the restriction $z > \varepsilon$ appears.

2. *Example 3.3.* The displayed conclusion reads “we still obtain $\tilde{T}_n(s) \xrightarrow{\mathbb{P}} T_n(s)$ but no longer $T_n(s) \nearrow T(s)$ ” and ends with “the importance of the assumption $k/n \not\rightarrow 0$ ”; the intended statements are $T_n(s) \nearrow T(s)$ and the importance of $k/n \rightarrow 0$ (equivalently $u_n \rightarrow \infty$).
3. *Proposition 3.2 (moment estimator).* In the decomposition of $M_n^{(2)}$ the third term should carry a minus sign, $-2 \log(Z_{n,n-k}/u_n) \int_{Z_{n,n-k}/u_n}^{\infty} \tilde{T}_n(t) t^{-1} dt$; see Exercise 9, (5). The term is $o_{\mathbb{P}}(1)$ either way. In the same proof the first term is in fact $o_{\mathbb{P}}(1)$, not merely bounded in probability.
4. *Exercise 10.* For the conclusion to hold the exponent must satisfy $\beta > 1/\gamma$ (as in Proposition 3.1); otherwise the remainder $O(t^{-\beta})$ dominates $c \log(t) t^{-1/\gamma}$ and F need not be regularly varying with index $1/\gamma$.